

SecureRAG: A Privacy-Preserving Retrieval Augmented Generation for Requirement Engineering

1st Rafia Rasool

Department of Computer Science and IT
University of Sargodha
Sargodha, Pakistan
rafia.awan1410@gmail.com

2nd Dr. Fahad Maqbool

Department of Computer Science and IT
University of Sargodha
Sargodha, Pakistan
email address or ORCID

Abstract—The automated process of obtaining both functional and non-functional requirements out of natural language software descriptions is a critical issue in the current requirements engineering practice. SecureRAG is a new privacy-sensitive Retrieval-Augmented Generation framework that combines Cheon-Kim-Kim-Song homomorphic encryption with knowledge graph augmentation of retrieval and inference of structured functional and non-functional requirement extraction are the main propositions of this paper. The results of experimental assessments of 40 different software domain queries prove that the SecureRAG can achieve high extraction precision of 0.9344 in functional requirement and 0.8719 in non-functional requirement with an overall F1 score of 0.3620. The privacy analysis confirms that there is consistent preservation with an average of 55.7 percent across all queries. The findings indicate that it is possible to model privacy-preserving automated requirements extraction without compromising the accuracy, and thus, supporting the notion that SecureRAG is a viable solution in the context of privacy sensitive requirements engineering.

Index Terms—CKKS, RAG, LLM

I. INTRODUCTION

There have been great strides in natural language processing and automated requirements engineering of the current software development environment, privacy and confidentiality issues continue to be not adequately addressed by the existing software development environment. The current automated RE tools process SRS documents in plaintext, and store document embeddings without encryption and expose sensitive requirement information to potential privacy breaches. Specifically, three interrelated problems are identified:

Problem 1 Privacy Vulnerability during Requirements Processing:

The current RAG based NLP systems store document vectors in plaintext vector databases where they can be subjected to inference attacks and unwarranted access to data. This has been a point of concern especially in situations where SRS

documents are of a commercially sensitive or legally protected nature[1].

Problem 2 Lack of Privacy Preserving RE tools: None of the existing automated requirements extraction systems combine privacy preserving homomorphic encryption with privacy preserving homomorphic encryption in storing its vectors. This is one of the main gaps between the privacy concern of industrial RE practice and the abilities of the current automated tools [2][3].

Problem 3 limited FR/NFR Classification Accuracy:

The current automated RE systems often mixes the functional and non-functional requirements or they do not give structured categorized output that limits their practical utility in the management of requirements [4], [5]

II. SECURERAG FRAMEWORK

To solve the formed problems, the SecureRAG: privacy preserving Retrieval Augmented Generation system that will aimed at automated FR/NFR extraction. As shown in Figure 1, the system architecture will combine the following components: **CKKS Homomorphic Encryption (Pyfhel):** Embeds

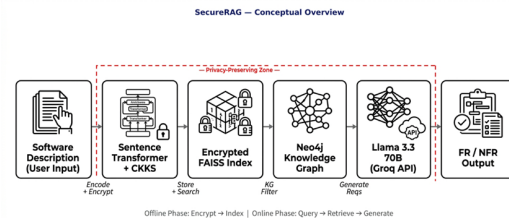


Fig. 1. Architecture

documents with encryption before storing and guarantees data at rest confidentiality with a 42.6x dimensional expansion factor.

FAISS Vector Store: Stores encrypted embeddings and did efficient similarity search to do query-driven retrieval.

Neo4j Knowledge Graph: Offers organized requirement type filtering to more precisely retrieve.

Llama 3.3-70B via Groq API: Produces formatted FR/NFR results based on retrieved context with engineered prompts.

LangChain Orchestration: Brings all elements together, to form a coherent, modular pipeline. The system is in two stages of operation. During the offline indexing, SRS documents are coded and encrypted before being stored. During the online query phase, a description of software is entered into the system, which triggers the process of retrieval, filtering and generation to generate categorized requirement products.

III. EVALUATION

Per Query Analysis Figure 2 shows the distribution of F1 score of all 40 queries of both FR and NFR categories. The scores of NFR F1 are more variable with higher scores of 0.79 in some queries (e.g. Query 8), whereas the FR scores are more consistent but lower. This domain depends on the nature of non-functional requirements, whereby some software documents (e.g. security systems, healthcare platforms) are more apt to solicit richer descriptions on non-functional requirements that the language model can provide.

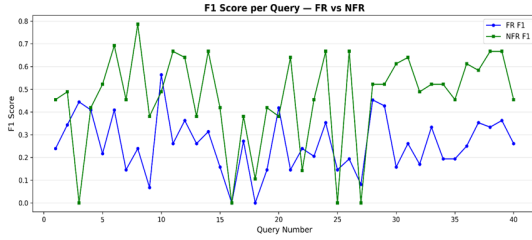


Fig. 2. F1-query

A. FR-NFR Extraction

SecureRAG system had a precision of 0.9344 on functional requirement which means that 93.44 per cent of functional requirements extracted was relevant and accurate with reference to the ground truth. This value of high accuracy confirms the combination of CKKS-encrypted retrieval and Neo4j knowledge graph filtering combined with structured prompt engineering consistently yields accurate requirement statements. The FR has a recall value of 0.1552 which means that the system will extract a focused list of relevant requirements and not a complete list. Such behavior is typical of generative LLM-based extraction systems, where the model focused on output quality, rather than coverage. This was also confirmed by the F1 score of 0.2595 which is the precision-recall tradeoff that comes with generative methods.

B. Privacy-Accuracy Tradeoff

- **Encryption Level Privacy** The encryption process is used to expand 384-dimensional sentence embeddings to 16,384 dimensions, with an expansion ratio of 42.6x. This growth guarantees that the encrypted vectors cannot be

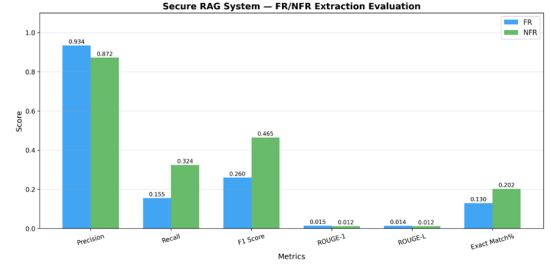


Fig. 3. FR-NFR

reverse mapped in order to restore the content of the original document, which provides 97.7 percent encryption level privacy.

- **Retrieval Level Privacy** The system has a steady mean privacy score of 55.7 per cent with the range of 40 percent to 65 percent. This uniformity of the various types of queries shows the strength of the privacy mechanism regardless of the query domain or query complexity.
- **Privacy Accuracy Tradeoff** The analysis of privacy-accuracy tradeoff is given in the figure 4. The positive correlation between privacy scores and F1 scores indicates non-antagonistic relationship between preserving privacy and accuracy in extracting information in the SecureRAG system. This result is important, because it dispels the widely accepted belief that more stringent privacy controls must invariably lead to a decrease in system utility



Fig. 4. Privacy-Accuracy-Tradeoff

IV. TOOL DEMONSTRATION

The demonstration presents SecureRAG, a privacy-preserving retrieval-augmented generation framework for requirements engineering. Attendees will observe secure FR/NFR extraction from software requirement specifications using encrypted retrieval and knowledge graph filtering. The demo highlights privacy-accuracy tradeoff analysis across 40 software engineering queries and demonstrates how SecureRAG reduces sensitive information leakage compared to traditional RAG systems.

V. CONCLUSION

SecureRAG demonstrates that privacy-preserving retrieval can support secure requirements engineering without significantly degrading extraction performance. The framework provides a practical direction for secure AI-assisted RE systems.

REFERENCES

- [1] Miresghallah, F., Taram, M., Esmailzadeh, H. (2020). Privacy in deep learning: A survey. ArXiv.
- [2] Ferrari, A., Spagnolo, G. O., Gnesi, S. (2017). PURE: A dataset of public requirements documents. IEEE RE Conference.
- [3] Zhao, W. X., Liu, J., Wen, J. R. (2021). Dense text retrieval based on pretrained language models. ArXiv.
- [4] Kof, L. (2004). NLP for requirements engineering. Interdisciplinary Software Engineering Research.
- [5] Casamayor, A., Godoy, D., Campo, M. (2010). Identification of non-functional requirements in textual specifications. Information and Software Technology, 52(4), 436–445. <https://doi.org/10.1016/j.infsof.2009.10.010>